R Package: Skimr

A quick look at any dataset

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Agenda:

What is skimr

How it works

Example dataset

What is skimr?

When we first open a dataset, we usually need to answer basic questions:

- Are any values missing?
- What are the typical values?
- How much do values vary?
- Are there possible unusual values?
- What does the distribution look like?

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skimr summarizes a dataset quickly

It gives a compact overview of variables so we can understand a dataset before cleaning, modeling, or creating visualizations.

One function → quick overview of your data

Why is it useful?

It saves time during exploratory data analysis

Without skimr, we may run many separate commands just to get basic information about one variable.

Without skimr → **With skimr**

```
mean(evals$score)
sd(evals$score)
min(evals$score)
median(evals$score)
max(evals$score)
```

```
skim(evals)
```

This connects to the basic summary statistics we use in class, but it makes the first step faster.

Example Dataset: Evals

100233 100004 102008 104000 100000 101005 100006 100603 100207 100233 100004
100603 104000 100026 102005 100027 100056 100000 100610 103029 100902 106007
100809 108000 103005 109020 100800 101077 100553 100669 100012 100834 100305
100262 130007 109001 104007 100303 102009 100000 100443 100202 100914 102006

	Course	score	rank	ethnicity	gender	language	age	cls_perc_eval
1	1	4.7	tenure track	minority	female	english	36	55.81395
2	2	4.1	tenure track	minority	female	english	36	68.80000
3	3	3.9	tenure track	minority	female	english	36	60.80000
4	4	4.8	tenure track	minority	female	english	36	62.60163
5	5	4.6	tenured	not minority	male	english	59	85.00000
6	6	4.3	tenured	not minority	male	english	59	87.50000

	cls_did_eval	cls_students	cls_level	cls_profs	bty_avg	pic_outfit	pic_color
1	24	43	upper	single	5	not formal	color
2	86	125	upper	single	5	not formal	color
3	76	125	upper	single	5	not formal	color
4	77	123	upper	single	5	not formal	color
5	17	20	upper	multiple	3	not formal	color
6	35	40	upper	multiple	3	not formal	color

RStudio Demo

How to use skimr?

```
# Load data
evals <- read.csv("/Users/admin/Desktop/DOC/DSS 665/evals.csv", header =
head(evals)

# Load skimr
library(skimr)

# Main function
skim(evals)
```

Returns:

- data summary and variable types
- missing values and complete rates
- mean, standard deviation, percentiles, min and max
- small distribution charts for numeric variables

With one command, skim() returns an organized summary by variable type.

Example 1: Full dataset overview

>skim(evals)

```
── Data Summary ────────────────────────
                           Values
Name                       evals
Number of rows             463
Number of columns          15
_________________________
Column type frequency:
  character                8
  numeric                  7
_________________________
Group variables            None
```

This is useful before analysis because I know the size and basic structure of the data immediately.

Example 2: Focus on variable(s)

```
>skim(evals
      )
```

```
── Variable type: character ──────────────────────────────────────────────────
  skim_variable n_missing complete_rate min max empty n_unique whitespace
1 rank                  0             1   7  12     0        3          0
2 ethnicity             0             1   8  12     0        2          0
3 gender                0             1   4   6     0        2          0
4 language              0             1   7  11     0        2          0
5 cls_level             0             1   5   5     0        2          0
6 cls_profs             0             1   6   8     0        2          0
7 pic_outfit            0             1   6  10     0        2          0
8 pic_color             0             1   5  11     0        2          0

── Variable type: numeric ────────────────────────────────────────────────────
  skim_variable n_missing complete_rate   mean      sd    p0    p25    p50   p75
1 Course                0             1 232    134.     1    116.   232    348.
2 score                 0             1   4.17   0.544  2.3    3.8    4.3    4.6
3 age                   0             1  48.4    9.80  29     42     48     57
4 cls_perc_eval         0             1  74.4   16.8   10.4   62.7   76.9   87.2
5 cls_did_eval          0             1  36.6   45.0    5     15     23     40
6 cls_students          0             1  55.2   75.1    8     19     29     60
7 bty_avg               0             1   4.42   1.53   1.67   3.17   4.33   5.5
```

```
>skim(evals, score)
```

```
── Variable type: numeric ──────────────────────────────────────────────────────
  skim_variable n_missing complete_rate mean    sd  p0 p25 p50 p75 p100 hist
1 score                 0             1 4.17 0.544 2.3 3.8 4.3 4.6    5 ▁▂▅▇
```

Example 3: Select variables + simplify output

> skim(evals, score, age, bty_avg)

```
── Variable type: numeric ──────────────────────────────────────────────────────
  skim_variable n_missing complete_rate  mean    sd    p0   p25   p50  p75  p100
1 score                 0             1  4.17 0.544  2.3   3.8   4.3   4.6  5
2 age                   0             1 48.4  9.80  29    42    48    57   73
3 bty_avg               0             1  4.42 1.53   1.67  3.17  4.33  5.5  8.17
  hist
1 ▁▃▇
2 ▃▇▃
3 ▇▅▁
```

> skim_without_charts(evals, score, age, bty_avg)

```
── Variable type: numeric ──────────────────────────────────────────────────────
  skim_variable n_missing complete_rate  mean    sd    p0   p25   p50  p75  p100
1 score                 0             1  4.17 0.544  2.3   3.8   4.3   4.6  5
2 age                   0             1 48.4  9.80  29    42    48    57   73
3 bty_avg               0             1  4.42 1.53   1.67  3.17  4.33  5.5  8.17
```

Short Summary

1. Quick overview
2. Flexible
3. Useful first step

Thank You

Sources

CRAN — Using skimr
https://cran.r-project.org/web/packages/skimr/vignettes/using_skimr.html

rOpenSci — skim(): Summarize a data frame or vector
https://docs.ropensci.org/skimr/reference/skim.html
https://ropensci.org/blog/2017/07/11/skimr/

Dataset: evals.csv (Canvas)